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Course Code & Name: 417522 – Data Modeling and Visualization	Class: BE	Sem-I
Course Coordinator: Mrs.D.A.Gore	Academic Year: 2024-25	

Overview of Subject:

This subject bridges the gap between raw data and actionable insights by focusing on two key areas: data modeling and data visualization.

Data Modeling

Data modeling involves the process of creating a conceptual representation of the data and its relationships. It is a fundamental step in understanding and organizing data to make it useful for analysis and decision-making. Key components of data modeling include:

- Understanding Data Requirements:**
 - Identify the types of data needed.
 - Determine the sources of data.
 - Define the **purpose** and scope of data usage.
- Data Characteristics Analysis:**
 - Analyze the properties of data, such as volume, variety, velocity, and veracity.
 - Assess the quality and reliability of data.
- Selecting Data Models:**
 - Choose appropriate data models such as relational, hierarchical, network, or object-oriented models based on data characteristics and requirements.
 - Develop schemas and data dictionaries to describe the structure and constraints of data.
- Probabilistic Data Modeling:**
 - Design models that incorporate uncertainty and probabilistic elements.
 - Use statistical methods to predict and interpret data patterns.
- Time Series Analysis:**
 - Evaluate data that is collected over time.
 - Use methods like trend analysis, seasonal decomposition, and forecasting to analyze time-dependent data.

Data Visualization

Data visualization is the graphical representation of data to help stakeholders understand complex data sets and derive actionable insights. Effective visualization techniques can transform abstract data into intuitive visual stories. Key components of data visualization include:

- Principles of Visualization:**

- Understand the principles of effective visualization, such as clarity, accuracy, and efficiency.
- Apply best practices in design to avoid misleading representations.
- 2. **Visualization Tools and Techniques:**
 - Use modern visualization tools such as Tableau, Power BI, matplotlib, and D3.js.
 - Implement various visualization techniques, including charts, graphs, maps, and dashboards.
- 3. **Exploratory Data Analysis (EDA):**
 - Use visualization as a tool for EDA to uncover patterns, trends, and anomalies.
 - Employ techniques like scatter plots, histograms, box plots, and heat maps to explore data visually.
- 4. **Data Cleaning and Transformation:**
 - Prepare data for visualization by loading, cleaning, transforming, merging, and reshaping it.
 - Handle missing values, outliers, and data inconsistencies.
- 5. **Communicating Insights:**
 - Present data findings effectively to different audiences.
 - Create visual stories that highlight key insights and support decision-making processes.

Unit 1: Introduction to Data Modelling (06 Hours)

Basic probability: Discrete and continuous random variables, independence, covariance, central limit theorem, Chebyshev inequality, diverse continuous and discrete distributions.

Statistics, Parameter Estimation, and Fitting a Distribution: Descriptive statistics, graphical statistics, method of moments, maximum likelihood estimation

Data Modeling Concepts • Understand and model subtypes and supertypes • Understand and model hierarchical data • Understand and model recursive relationships • Understand and model historical data

Session No. 1

Basic probability:

Discrete and continuous random variables

Real Time Analogy

Discrete random variable

Imagine you're playing a game where you roll a six-sided die. The outcome of this roll can only be one of six discrete possibilities: 1, 2, 3, 4, 5, or 6. Each outcome is distinct and separate from the others, and you can count the exact number of times each outcome occurs. This is analogous to a discrete random variable, where the possible values are countable and separate.

Continuous random variables

Now, imagine instead of rolling a die, you're measuring the temperature outside. The

temperature can take on any value within a certain range (say, between 0 to 100 degrees Celsius), and it can be any value within that range, not just specific points like with the die. For example, it could be 23.5°C, 24.1°C, or any other decimal value within the range. The temperature can vary continuously without distinct separation between values. This is analogous to a continuous random variable, where the possible values form a continuous spectrum.

Real-Time Application of Discrete Random Variables

1. Call Center Queue: The number of incoming calls to a customer service center in a minute can be modeled as a discrete random variable. Each call represents a discrete event that can be counted.
2. Number of Emails Received: The count of emails received in an hour by an individual or organization is another example. Each email arrival is a separate event that can be quantified.

Real-Time Application of Continuous Random Variables

1. Temperature Measurement: Continuous random variables are used to represent measurements like temperature, which can take any value within a range. For example, measuring the temperature every second gives a continuous stream of data points.
2. Stock Prices: The price of a stock in financial markets is a continuous random variable. Stock prices change continuously throughout trading hours and can take on any real number value.

What is Random Variable in Statistics?

In probability, a real-valued function, defined over the sample space of a random experiment, is called a random variable. That is, the values of the random variable correspond to the outcomes of the random experiment. Random variables could be either discrete or continuous. In this article, let's discuss the different types of random variables.

Random Variable Definition

A random variable is a rule that assigns a numerical value to each outcome in a sample space. Random variables may be either discrete or continuous. A random variable is said to be discrete if it assumes only specified values in an interval. Otherwise, it is continuous. We generally denote the random variables with capital letters such as X and Y. When X takes values 1, 2, 3, ..., it is said to have a discrete random variable.

Variate

A variate can be defined as a generalization of the random variable. It has the same properties as that of the random variables without stressing to any particular type of probabilistic experiment. It always obeys a particular probabilistic law.

Types of Random Variable

As discussed in the introduction, there are two random variables, such as:

- Discrete Random Variable
- Continuous Random Variable

Random Variable Formula

For a given set of data the mean and variance random variable is calculated by the formula. So, here we will define two major formulas:

Mean of random variable

Variance of random variable

Mean of random variable: If X is the random variable and P is the respective probabilities, the mean of a random variable is defined by:

$$\text{Mean } (\mu) = \sum XP$$

where variable X consists of all possible values and P consists of respective probabilities.

Variance of Random Variable: The variance tells how much is the spread of random variable X around the mean value. The formula for the variance of a random variable is given by,

$$\text{Var}(X) = \sigma^2 = E(X^2) - [E(X)]^2$$

$$\text{where } E(X^2) = \sum X^2P \text{ and } E(X) = \sum XP$$

Random Variable and Probability Distribution

The probability distribution of a random variable can be

Theoretical listing of outcomes and probabilities of the outcomes.

An experimental listing of outcomes associated with their observed relative frequencies.

A subjective listing of outcomes associated with their subjective probabilities.

The probability of a random variable X which takes the values x is defined as a probability function of X is denoted by $f(x) = f(X=x)$

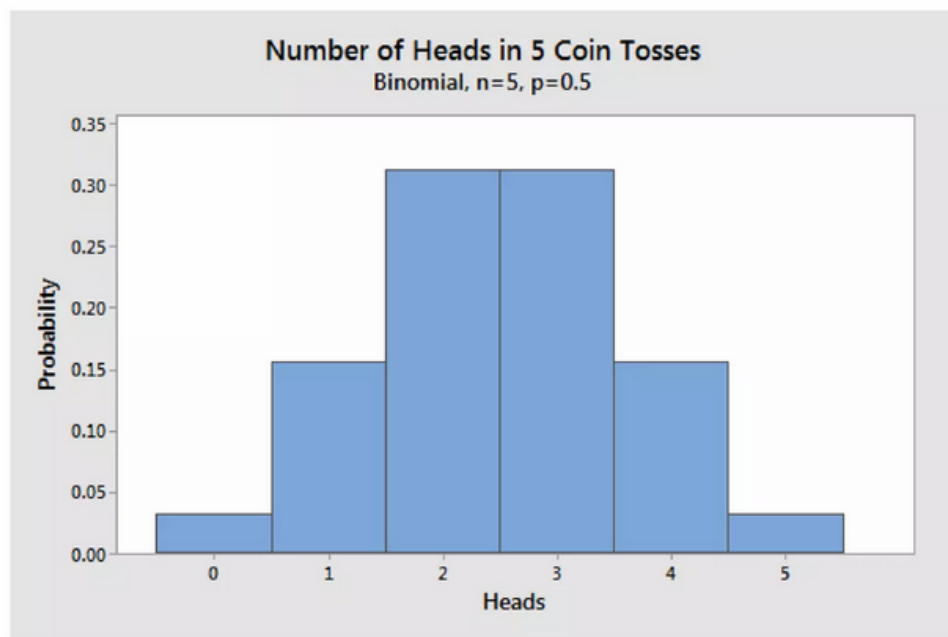
A probability distribution always satisfies two conditions:

The important probability distributions are:

- Binomial distribution
- Poisson distribution
- Bernoulli's distribution
- Exponential distribution
- Normal distribution

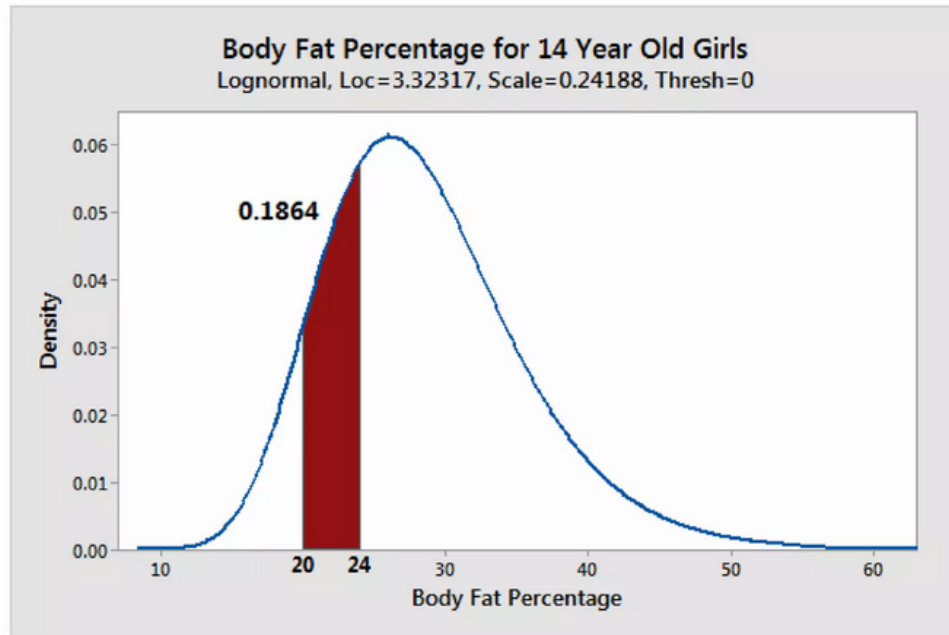
Discrete Example

The number of heads that appear during a series of five coin tosses is a discrete random variable that follows the binomial distribution. We can use that distribution to determine the likelihood of obtaining 0 to 5 heads. The graph below displays the probability for each possible outcome.



Continuous Example

Body fat percentage is a continuous random variable. In preteen girls, it follows a lognormal distribution. Suppose a researcher needs to find participants with a body fat percentage between 20 and 24 percent. What is the likelihood that the next candidate will fall within that range?



The lognormal distribution graph indicates that the probability of body fat falling in the range of 20 to 24% is 0.1864. This information can help the researcher determine how many candidates they'll need to assess to obtain a sufficient sample size.

Independence

Independence is a fundamental notion in probability theory, as in statistics and the theory of stochastic processes. Two events are independent, statistically independent, or stochastically independent if, informally speaking, the occurrence of one does not affect the probability of occurrence of the other or, equivalently, does not affect the odds. Similarly, two random variables are independent if the realization of one does not affect the probability distribution of the other.

When dealing with collections of more than two events, two notions of independence need to be distinguished. The events are called pairwise independent if any two events in the collection are independent of each other, while mutual independence (or collective independence) of events

Covariance

Real-Time Analogy of Covariance

Imagine you are monitoring two financial assets in real-time, let's say the stock prices of two companies, Company A and Company B. Covariance, in this context, measures how the prices of these two stocks move together over time. Here's how you can relate covariance to this scenario:

1. **Stock Price Movements:** Suppose you have historical data of the minute-by-minute changes in stock prices for both companies. Each data point represents the price of Company A's stock and Company B's stock at a specific time.
2. **Covariance Calculation:** Covariance between the stock prices of Company A and

Company B measures the tendency of their prices to move together. If the prices of both stocks tend to increase or decrease together consistently, their covariance would be positive. If one tends to increase when the other decreases, their covariance would be negative.

Real-Time Application of Covariance

Portfolio Risk Management: In finance, covariance is used to measure the risk associated with holding multiple assets in a portfolio. A portfolio manager needs to understand how the returns of different assets move together. Positive covariance indicates that the returns tend to move in the same direction, while negative covariance suggests they move in opposite directions. By considering covariance, investors can construct diversified portfolios to minimize risk.

Covariance Meaning

Covariance is a measure of the relationship between two random variables and to what extent, they change together. Or we can say, in other words, it defines the changes between the two variables. such that change in one variable is equal to change in another variable. This is the property of a function of maintaining its form when the variables are linearly transformed. Covariance is measured in units, which are calculated by multiplying the units of the two variables.

Types of Covariance

Covariance can have both positive and negative values. Based on this, it has two types:

1. Positive Covariance
2. Negative Covariance

Positive Covariance

If the covariance for any two variables is positive, that means, both the variables move in the same direction. Here, the variables show similar behaviour. That means, if the values (greater or lesser) of one variable corresponds to the values of another variable, then they are said to be in positive covariance.

Negative Covariance

If the covariance for any two variables is negative, that means, both the variables move in the opposite direction. It is the opposite case of positive covariance, where greater values of one variable correspond to lesser values of another variable and vice-versa.

Covariance Formula

Covariance formula is a statistical formula, used to evaluate the relationship between two variables. It is one of the statistical measurements to know the relationship between the variance

between the two variables. Let us say X and Y are any two variables, whose relationship has to be calculated. Thus the covariance of these two variables is denoted by $\text{Cov}(X,Y)$. The formula is given below for both population covariance and sample covariance.

Population Covariance Formula

$$\text{Cov}(x,y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N}$$

Sample Covariance

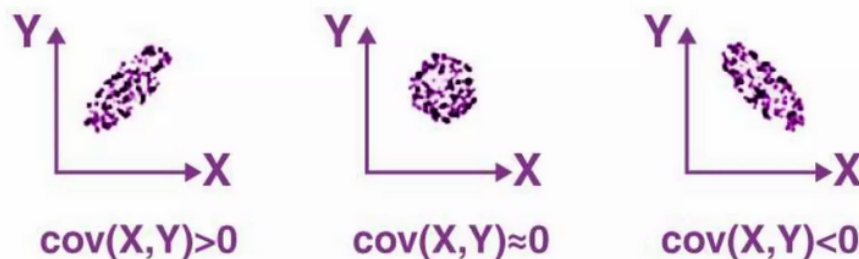
$$\text{Cov}(x,y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N - 1}$$

where,

- x_i = data value of x
- y_i = data value of y
- $\bar{x}$ = mean of x
- $\bar{y}$ = mean of y
- N = number of data values.

Covariance of X and Y

Below figure shows the covariance of X and Y.



If $\text{cov}(X, Y)$ is greater than zero, then we can say that the covariance for any two variables is positive and both the variables move in the same direction.

If $\text{cov}(X, Y)$ is less than zero, then we can say that the covariance for any two variables is negative and both the variables move in the opposite direction.

If $\text{cov}(X, Y)$ is zero, then we can say that there is no relation between two variables.

Correlation Coefficient Formula

We have already discussed covariance, which is the evaluation of changes between any two variables. Correlation estimates the depth of the relationship between variables. It is the estimated measure of covariance and is dimensionless. In other words, the correlation coefficient is a constant value always and does not have any units. The relationship between the correlation coefficient and covariance is given by,

$$\text{Correlation, } \rho(X,Y) = \text{Cov}(X,Y) / \sigma_X \sigma_Y$$

Where:

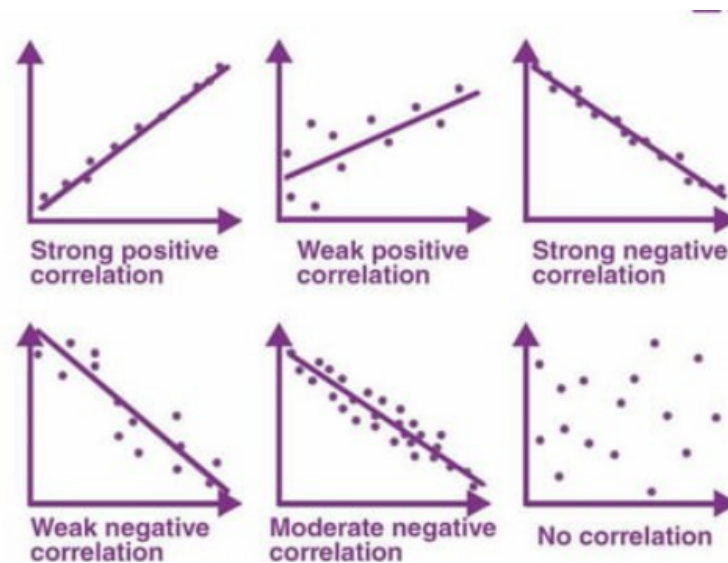
$\rho(X,Y)$ correlation between the variables X and Y

$\text{Cov}(X,Y)$ covariance between the variables X and Y

σ_X standard deviation of the X variable

σ_Y standard deviation of the Y variable

Based on the value of correlation coefficient, we can estimate the type of correlation between the given two variables. Also, the graphical representation of correlation among two variables is given in the below figure.



Covariance and Correlation

Below table shows the comparison among covariance and correlation in brief.

Covariance	Correlation
It is a measure to show the extent to which random variables change with respect to each other, given two random variables are related to each other	It is a measure used to describe how strongly the er. given two random variables are related to each other
It is a measure of correlation.	It is defined as the scaled form of covariance,
The value of covariance lies between $-\infty$ and $+\infty$	The value of correlation lies between -1 and +1.
It indicates the direction of the linear relationship between the given two variables.	It measures the direction and strength of the linear relationship between the given two variables.

Covariance and Variance

Covariance and variance both are the terms used in statistics. Variance is the measure of spread of data around its mean value but covariance measures the relation between two random variables.

Covariance Example

Below example helps in better understanding of the covariance of two variables.

Question:

Calculate the coefficient of covariance for the following data:

X	2	8	18	20	28	30
Y	5	12	18	23	45	50

Solution:

Number of observations = 6

Mean of X=17.67

Mean of Y=25.5

$$\text{Cov}(X, Y) = (1/6) [(2 - 17.67)(5 - 25.5) + (8 - 17.67)(12 - 25.5) + (18 - 17.67)(18 - 25.5) + (20 - 17.67)(23 - 25.5) + (28 - 17.67)(45 - 25.5) + (30 - 17.67)(50 - 25.5)]$$

$$= 157.83$$

Session No. 2

Central limit theorem

Real-Time Analogy of Central Limit Theorem

Imagine you are conducting an experiment where you measure the weights of randomly selected apples from different trees in an orchard. Each time you select a batch of apples and calculate their average weight, you record the result. Here's how the Central Limit Theorem applies to this scenario:

1. **Sampling Distribution:** Each batch of apples represents a sample from the population of all apples in the orchard. According to the Central Limit Theorem, regardless of the shape of the individual apple weights' distribution, the distribution of the sample means (average weights of the batches) will tend to follow a normal distribution as the sample size increases.
2. **Real-Time Experiment:** As you continue to take more samples (batches of apples) and compute their average weights, you notice that the distribution of these sample means (average weights) starts to resemble a bell-shaped curve, even if the weights of

individual apples are not normally distributed.

Real-Time Applications of Central Limit Theorem

1. Quality Control in Manufacturing

Scenario: A manufacturing plant produces electronic components, and quality control involves measuring the dimensions of randomly selected components from each batch.

- **Application:** According to the CLT, the distribution of sample means of component dimensions will tend to follow a normal distribution, regardless of the individual dimensions' distributions. This allows quality inspectors to estimate the population mean and assess whether the batch meets specifications based on sample data.

2. Polling and Survey Analysis

Scenario: A polling agency conducts surveys to estimate public opinion on political candidates. Each survey randomly selects a sample of respondents.

- **Application:** The CLT ensures that the distribution of sample means of survey responses (such as approval ratings) will approximate a normal distribution. This enables pollsters to calculate margins of error and confidence intervals for their estimates, providing reliable predictions despite variations in individual survey responses.

The central limit theorem, which is a statistical theory, states that when a large sample size has a finite variance, the samples will be normally distributed, and the mean of samples will be approximately equal to the mean of the whole population.

In other words, the central limit theorem states that for any population with mean and standard deviation, the distribution of the sample mean for sample size N has mean μ and standard deviation

As the sample size gets bigger and bigger, the mean of the sample will get closer to the actual population mean. If the sample size is small, the actual distribution of the data may or may not be normal, but as the sample size gets bigger, it can be approximated by a normal distribution. This statistical theory is useful in simplifying analysis while dealing with stock indexes and much more.

Central Limit Theorem Statement

The central limit theorem states that whenever a random sample of size n is taken from any distribution with mean and variance, then the sample mean will be approximately a normal distribution with mean and variance.

Assumptions of the Central Limit Theorem

- The sample should be drawn randomly following the condition of randomisation.
- The samples drawn should be independent of each other. They should not influence the other samples.

- When the sampling is done without replacement, the sample size shouldn't exceed 10% of the total population.
- The sample size should be sufficiently large.

Formula

The formula for the central limit theorem is given below:

Central Limit Theorem for Sample Means,

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Examples on Central Limit Theorem

Example 1:

20 students are selected at random from a clinical psychology class; find the probability that their mean GPA is more than 5. If the average GPA scored by the entire batch is 4.91, the standard deviation is 0.72.

Solution:

Here,

Population mean = $\mu = 4.91$

Population standard deviation = $\sigma = 0.72$

Sample size = $n = 20$ (which is less than 30)

Since the sample size is smaller than 30, use the t-score instead of the z-score, even though the population standard deviation is known.

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

Substituting the values, we have

$$\begin{aligned}\sigma_{\bar{x}} &= \frac{0.72}{\sqrt{20}} \\ &= 0.161\end{aligned}$$

Now, find the t-score:

$$t = \frac{x - \mu}{\sigma_x}$$

For this problem, the raw score $x = 5$

$$\begin{aligned} t &= \frac{5 - 4.91}{0.161} \\ &= 0.559 \end{aligned}$$

Find the probability for the t value using the t-score table. The degree of freedom here would be:

$$Df = 20 - 1 = 19$$

$$P(t \leq 0.559) = 0.7087$$

$$P(t > 0.559) = 1 - 0.7087 = 0.2913$$

Applications of Central Limit Theorem

- 1] The sample distribution is assumed to be normal when the distribution is unknown or not normally distributed according to the central limit theorem. This method assumes that the given population is distributed normally. It helps in data analysis.
- 2] The sample mean deviation decreases as we increase the samples taken from the population, which helps in estimating the mean of the population more accurately.
- 3] The sample mean is used to create a range of values which likely includes the population mean.
- 4] The concept of the central limit theorem is used in election polls to estimate the percentage of people supporting a particular candidate as confidence intervals.
- 5] CLT is used in calculating the mean family income in a particular country.
- 6] It is used in rolling many identical, unbiased dice.
- 7] The probability distribution for the total distance covered in a random walk will approach a normal distribution.
- 8] Flipping many coins will result in a normal distribution for the total number of heads (or, equivalently total number of tails).
- 9] By looking at the sample distribution, CLT can tell whether the sample belongs to a particular population.
- 10] It enables us to make conclusions about the sample and population parameters and assists in constructing good machine-learning models.

Chebyshev inequality

In plain English, the formula says that the probability of a value (X) being more than k standard deviations () away from the mean () is less than or equal to . In that formula, it represents a distance away from the mean expressed in standard deviation units, where we can take any above-zero value.

As an example, let's calculate the probability for k=2.

$$Pr(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}$$

Although the inequality is valid for any real number $k > 0$, only the case $k > 1$ is practically useful.

As an example, let's calculate the probability for k=2.

$$Pr(|X - \mu| \geq 2 \times \sigma) \leq \frac{1}{4}$$

The answer is that no more than ¼ of values will be more than 2 standard deviations away from the distribution's mean. Note that this is a loose upper bound, but it is valid for any distribution.

Real Time Application

Assume that an asset is picked from a population of assets at random. The average return of the population of assets is 12%, and the standard deviation of the population of assets is 5%. To calculate the probability that an asset picked at random from this population, which has a return less than 4% or greater than 20%, Chebyshev's inequality can be applied.

Since there is a limited amount of information and only the mean and standard deviation of a distribution is given, the exact probability of this scenario cannot be determined; thus, Chebyshev's inequality is applied. Below is the application of the theory:

$$|X - \mu| \geq K$$

$$P(|X - \mu| \geq K) \leq (\sigma^2 / K^2) = (5\%^2 / 8\%^2)$$

$$P(|X - \mu| \geq K) \leq (\sigma^2 / K^2) = 39.06\%$$

Where:

- **Standard deviation:** 5%
- **Mean:** 12%
- **K:** 8%

Thus, the probability of an asset's return to be less than 4% or greater than 20% from the population of assets, which has a mean return of 12% with a standard deviation of 5%, is less than 39.06%, according to Chebyshev's inequality.

Diverse continuous and discrete distributions

What is Probability Distribution?

Probability Distribution is a statistical function which links or lists all the possible outcomes a random variable can take, in any random process, with its corresponding probability of occurrence.

Values of random variable changes, based on the underlying probability distribution.

It gives the idea about the underlying probability distribution by showing all possible values which a random variable can take along with the likelihood of those values.

Let X be the number of heads that result from the toss of 2 coins. Here X can take values 0, 1, or 2. X is a discrete random variable.

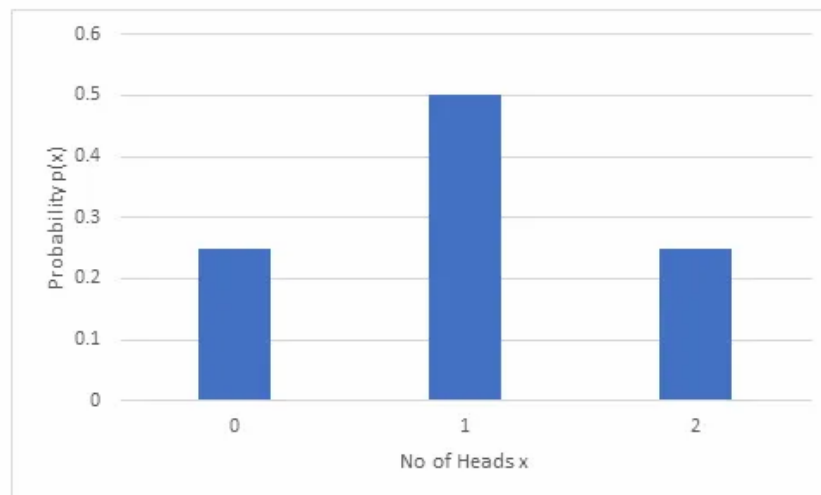
The table below shows the probabilities associated with the different possible values of X .

The probability of getting 0 heads is 0.25, 1 head is 0.50, 2 heads is 0.25.

Simple example of probability distribution for a discrete random variable.

No of Heads(X)	Probability $P(X)$
0	0.25
1	0.5
2	0.25

Probabilities distribution of random variable X



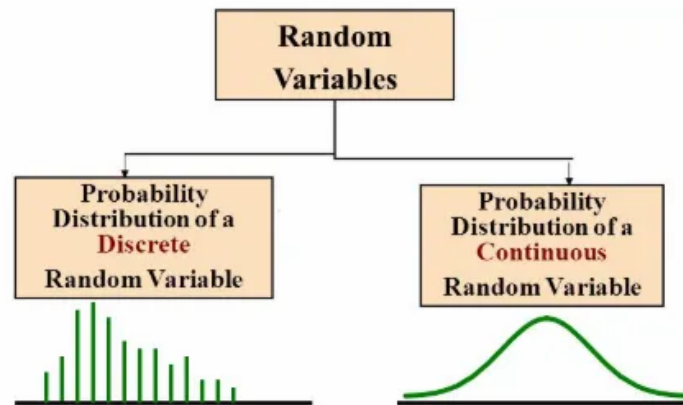
Probability Distribution for the above example

Need of Probability Distribution

According to the definition of a random variable, it's the variable which can hold a different set of values from the outcome of any random process. However, it lacks the capability to capture the probability of getting those different values.

So, probability distribution helps to create a clear picture of all the possible sets of values with their respective probability of occurrence in any random process.

Different Probability Distributions



Probability Distribution of Discrete and Continuous Random Variable.

If a random variable can take only a finite set of values (Discrete Random Variable), then its probability distribution is called a Probability **Mass Function or PMF**.

Probability Distribution of discrete random variables is the list of values of different outcomes and their respective probabilities.

In other words, for a discrete random variable X , the value of the Probability Mass Function $P(x)$ is given as,

$$P(x) = P(X=x)$$

If X , a discrete random variable takes different values $x_1, x_2, x_3, \dots$

Then,

$$\sum_{i=1}^{\infty} p(x_i) = 1.$$

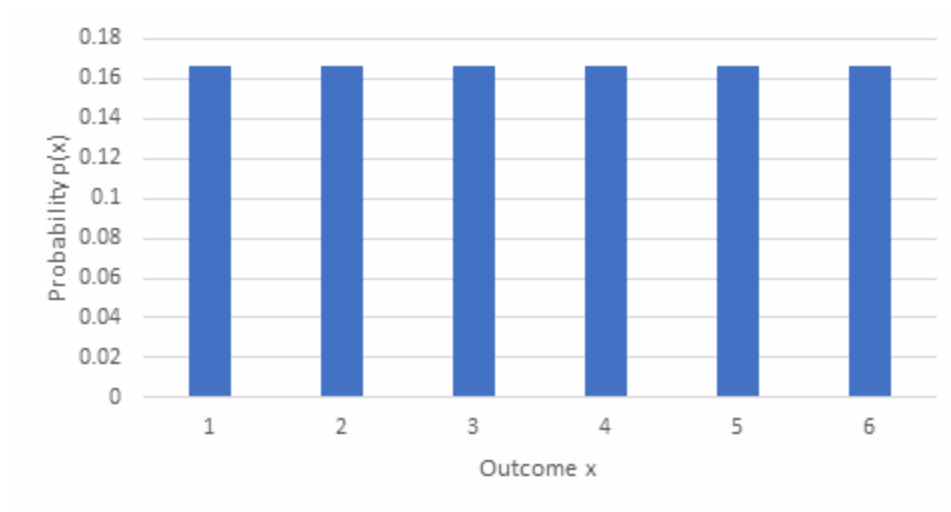
And $0 \leq p(x_i) \leq 1$

Example: Rolling of a Dice.

If X is a random variable associated with the rolling of a six-sided fair dice then, PMF of X is given as:

X	$x=1$	$x=2$	$x=3$	$x=4$	$x=5$	$x=6$
$P(X=x)$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$

PMF



Unlike discrete random variables, continuous random variables hold different values from an interval of real numbers. Hence it's difficult to sum these uncountable values like discrete random variables and therefore integral over those set of values is done.

Probability distribution of continuous random variables is called the Probability **Density function or PDF**.

Given the probability function $P(x)$ for a random variable X , the probability that X belongs to A , where A is some interval is calculated by integrating $p(x)$ over the set A i.e

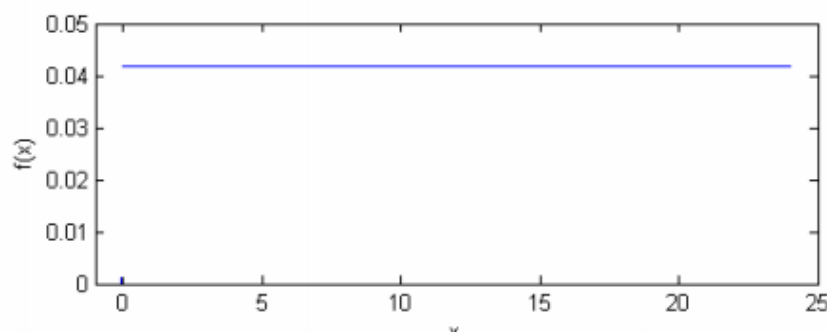
$$P(X \in A) = \int_A p(x) dx$$

Where, $0 \leq p(x) \leq 1$ for all x and $\int p(x) dx = 1$

Example: A clock stops at any random time during the day. Let X be the time (Hours plus fractions of hours) at which the clock stops. The PDF for X is

$$f(x) = \begin{cases} \frac{1}{24} & 0 \leq x \leq 24 \\ 0 & \text{otherwise} \end{cases}$$

And the density curve is given by



What is the difference between a discrete probability distribution and a continuous probability distribution?

- In discrete probability distributions, the random variable associated with it is discrete, whereas in continuous probability distributions, the random variable is continuous.
- Continuous probability distributions are usually introduced using probability density functions, but discrete probability distributions are introduced using probability mass functions.
- The frequency plot of a discrete probability distribution is not continuous, but it is continuous when the distribution is continuous.
- The probability that a continuous random variable will assume a particular value is zero, but it is not the case in discrete random variables.

Session No. 3

Statistics, Parameter Estimation and Fitting a Distribution

Real Time Application

Decision Support: By analyzing temperature data in real-time, stakeholders can make informed decisions related to weather forecasts, energy consumption management, agricultural planning, and infrastructure maintenance.

Continuous Monitoring: Real-time statistics and distribution fitting enable proactive responses to temperature fluctuations, helping to mitigate risks associated with extreme weather events or climate changes.

Forecasting and Predictive Analytics: Parameters estimated from fitted distributions can be used for forecasting future temperature trends, aiding in resource allocation and disaster preparedness.

Statistics:

Statistics is a branch of mathematics that deals with the collection, analysis, interpretation, presentation, and organization of data. It provides methods and techniques to extract meaningful information from data, make inferences, and draw conclusions about populations based on sample data. There are two main branches of statistics:

a. Descriptive Statistics: Descriptive statistics involve summarizing and presenting data in a meaningful way. Measures such as mean, median, mode, standard deviation, and percentiles are commonly used to describe the central tendency and variability of the data.

b. Inferential Statistics: Inferential statistics use sample data to make inferences or predictions about the larger population from which the sample was drawn. Techniques like hypothesis testing, confidence intervals, and regression analysis fall under inferential statistics.

Parameter Estimation:

Parameter estimation is a fundamental aspect of statistics that involves determining the unknown parameters of a statistical model based on observed data. In many real-world scenarios, data is generated from a specific statistical distribution with unknown parameters. The goal of parameter estimation is to find the best estimates for these unknown parameters, often denoted as θ , so that the statistical model closely represents the observed data.

There are two main approaches to parameter estimation

a. Method of Moments: In this approach, the parameters of the distribution are estimated by equating sample moments (such as the mean, variance, etc.) to the theoretical moments of the distribution.

b. Maximum Likelihood Estimation (MLE): MLE is a powerful and widely used method for parameter estimation. It involves finding the parameter values that maximize the likelihood function, which measures the probability of obtaining the observed data given a particular set of parameter values.

Statistical Models

Engineers and scientists apply parameter estimation to statistical models to estimate:

- Parameters of a probability distribution, such as the mean and standard deviation of a normal distribution
- Regression coefficients of a regression model.

The Statistics and Machine Learning Toolbox™ supports these and similar parameter estimation tasks with more than 40 different probability distributions, including normal, Weibull, gamma, generalized Pareto, and Poisson. The toolbox also supports linear and nonlinear regression.

Dynamic Models

Engineers apply parameter estimation to dynamic models to compute;

- Coefficients of transfer functions, including ARX, ARMAX, Box-Jenkins, and output-error models
- Entries in state-space matrices
- Coefficients of ODEs or well-structured systems with parameter constraints (grey-box system identification)
- Regression coefficients, saturation levels, or dead-zone limits for nonlinear dynamic systems, including nonlinear ARX and Hammerstein-Wiener

The System Identification Toolbox supports parameter estimation for linear and nonlinear parametric dynamic models.

Methods of Parameter Estimation

The techniques used for parameter estimation are called estimators.

Some estimators are:

- **Probability Plotting:** A method of finding parameter values where the data is plotted on special plotting paper and parameters are derived from the visual plot.
- **Rank Regression (Least Squares):** A method of finding parameter values that minimizes the sum of the squares of the residuals.
- **Maximum Likelihood Estimation:** A method of finding parameter values that, given a set of observations, will maximize the likelihood function.
- **Bayesian Estimation Methods:** A family of estimation methods that tries to minimize the posterior expectation of what is called the utility function. In practice, what this means is that existing knowledge about a situation is formulated, data is gathered, and then posterior knowledge is used to update our beliefs.

Fitting a Distribution:

Fitting a distribution is the process of selecting an appropriate probability distribution that best describes a given dataset. As mentioned earlier, this process involves choosing candidate distributions and estimating their parameters based on the observed data,

Common steps involved in fitting a distribution include:

- a. **Data Selection and Preprocessing:** Gather relevant data and prepare it for analysis by handling missing values, outliers, and transforming the data if necessary.
- b. **Candidate Distributions:** Select a set of candidate distributions based on the characteristics of the data and the underlying phenomenon.
- c. **Parameter Estimation.** Estimate the parameters of each candidate distribution using methods like the method of moment or maximum likelihood estimation,
- d. **Goodness of Fit Testing:** Evaluate how well each candidate distribution fits the data using goodness-of-fit tests or graphical methods.
- e. **Model Selection:** Choose the best-fitting distribution among the candidates based on the goodness-of-fit results.
- f. **Interpretation and Inference.** Use the selected distribution and estimated parameters for predictions, simulations, or gaining insights into the phenomenon.

Descriptive statistics

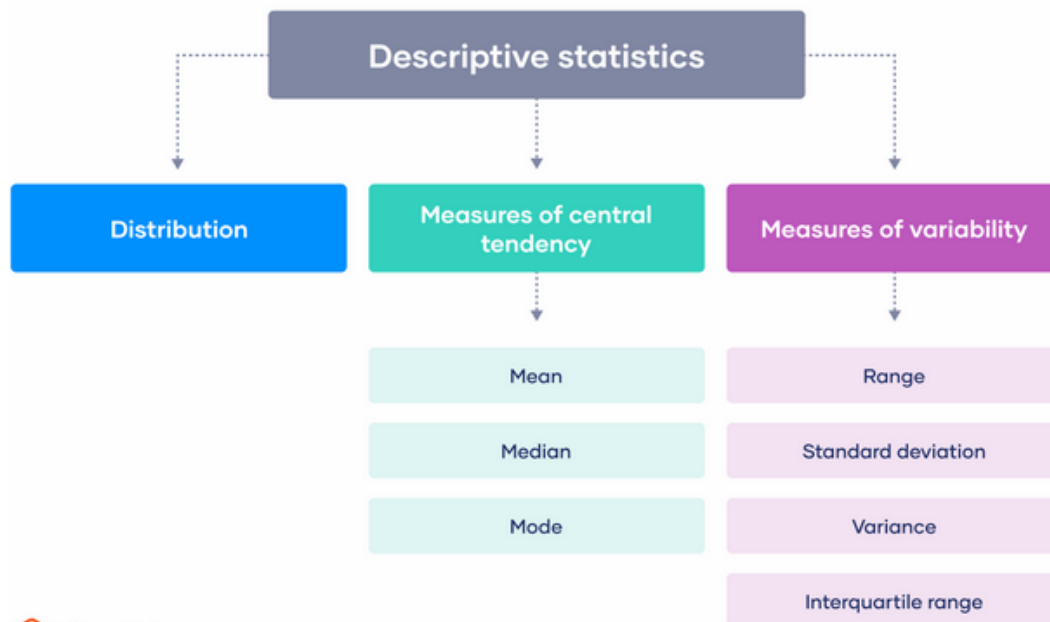
What Are Descriptive Statistics?

Descriptive statistics are brief informational coefficients that summarize a given data set, which can be either a representation of the entire population or a sample of a population. Descriptive statistics are broken down into measures of central tendency and measures of variability (spread). Measures of central tendency include the mean, median, and mode, while measures of variability include standard deviation, variance, minimum and maximum variables, kurtosis, and skewness.

Types of descriptive statistics

There are 3 main types of descriptive statistics:

- The distribution concerns the frequency of each value.
- The central tendency concerns the averages of the values.
- The variability or dispersion concerns how spread out the values are.



Distribution

Distribution (or frequency distribution) refers to the quantity of times a data point occurs. Alternatively, it is the measurement of a data point failing to occur. Consider a data set: male, male, female, female, female, other. The distribution of this data can be classified as:

- The number of males in the data set is 2.
- The number of females in the data set is 3.
- The number of individuals identifying as others is 1.
- The number of non-males is 4.

Univariate vs. Bivariate

In descriptive statistics, **univariate data** analyzes only one variable. It is used to identify characteristics of a single trait and is not used to analyze any relationships or causations.

For example, imagine a room full of high school students. Say you wanted to gather the average age of the individuals in the room. This univariate data is only dependent on one factor: each person's age. By gathering this one piece of information from each person and dividing by the total number of people, you can determine the average age.

Bivariate data, on the other hand, attempts to link two variables by searching for correlation. Two types of data are collected, and the relationship between the two pieces of information is

analyzed together. Because multiple variables are analyzed, this approach may also be referred to as multivariate.

Session No. 4

Graphical Statistics

Real-Time Analogy of Graphical Statistics

Scenario: Imagine you are responsible for monitoring traffic conditions in a city using real-time data from traffic cameras and sensors deployed across key intersections and highways.

1. Data Collection and Visualization:

- **Real-Time Data Collection:** Traffic cameras and sensors continuously capture data such as vehicle speeds, traffic volumes, and congestion levels at various points throughout the city.
- **Graphical Representation:** To monitor traffic conditions effectively, you use graphical statistics to visualize this data in real-time. Graphical techniques such as charts, graphs, and maps are employed to represent different aspects of traffic flow and congestion.

There are different types of graphs in mathematics and statistics which are used to represent data in a pictorial form. Among the various types of charts or graphs, the most common and widely used ones are explained below.

Types of Graphs and Charts

The list of most commonly used graph types are as follows:

- Statistical Graphs (bar graph, pie graph, line graph, etc.)
- Exponential Graphs
- Logarithmic Graphs
- Trigonometric Graphs

Frequency Distribution Graph

All these graphs are used in various places to represent a specific set of data concisely. The details of each of these graphs (or charts) are explained below in detail which will not only help to know about these graphs better but will also help to choose the right kind of graph for a particular data set

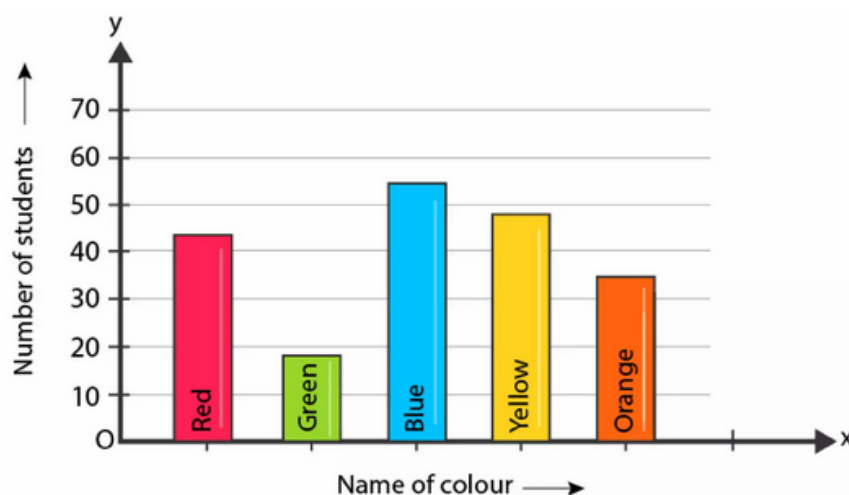
Statistical Graphs

A statistical graph or chart is defined as the pictorial representation of statistical data in graphical form. The statistical graphs are used to represent a set of data to make it easier to understand and interpret statistical information. The different types of graphs that are commonly used in statistics are given below.

Types of Graphs in Statistics

Bar Graph

Bar graphs are the pictorial representation of grouped data in vertical or horizontal rectangular bars, where the length of bars is proportional to the measure of data. The chart's horizontal axis represents categorical data, whereas the chart's vertical axis defines discrete data.



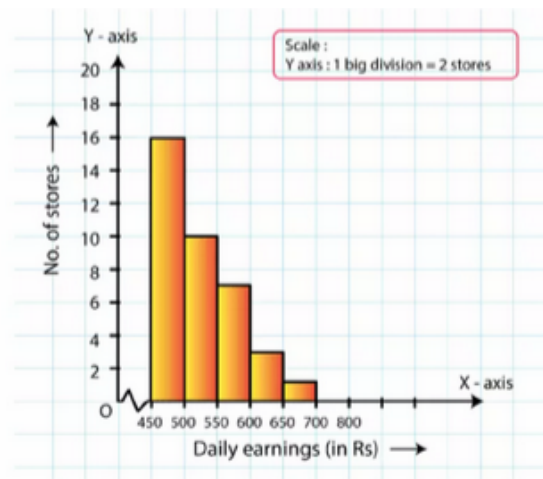
Line Graph

A graph that utilizes points and lines to represent change over time is defined as a line graph. In other words, it is a chart that shows a line joining several points or a line that shows the relation between the points. The diagram depicts quantitative data between two changing variables with a straight line or curve that joins a series of successive data points. Linear charts compare these two variables on a vertical and horizontal axis.



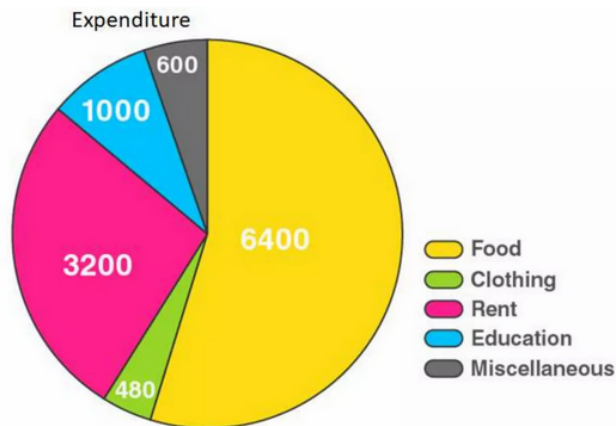
Histogram

A histogram chart displays the frequency of discrete and continuous data in a dataset using connected rectangular bars. Here, the number of observations that fall into a predefined class interval represented by a rectangular bar.



Pie Chart

A pie chart used to represent the numerical proportions of a dataset. This graph involves dividing a circle into various sectors, where each sector represents the proportion of a particular element as a whole. This is also called a circle chart or circle graph.

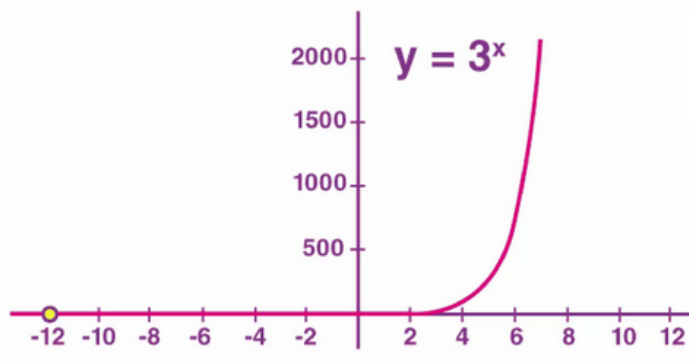


Exponential Graphs

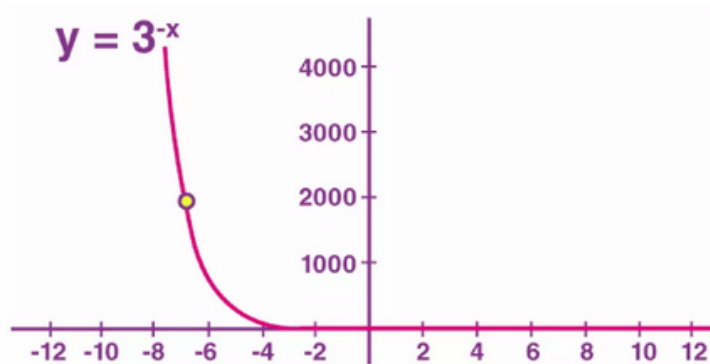
Exponential graphs are the representation of exponential functions using the table of values and plotting the points on a graph paper. It should be noted that the exponential functions are the inverse of logarithmic functions. In the case of exponential charts, the graph can be an increasing or decreasing type of curve based on the function. An example is given below, which will help to understand the concept of graphing exponential function easily.

For example, the graph of $y = 3^x$ is an increasing one while the graph of $y = 3^{-x}$ is a decreasing one.

Graph of $y = 3^x$:



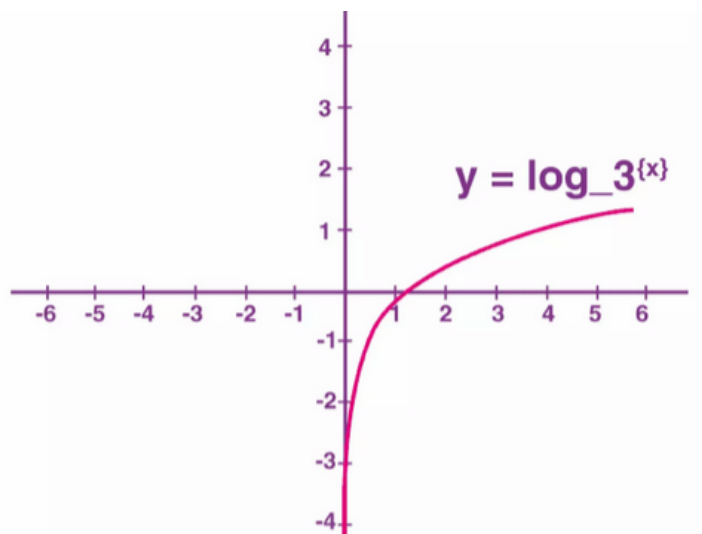
Graph of $y = 3^{-x}$:



Logarithmic Graphs

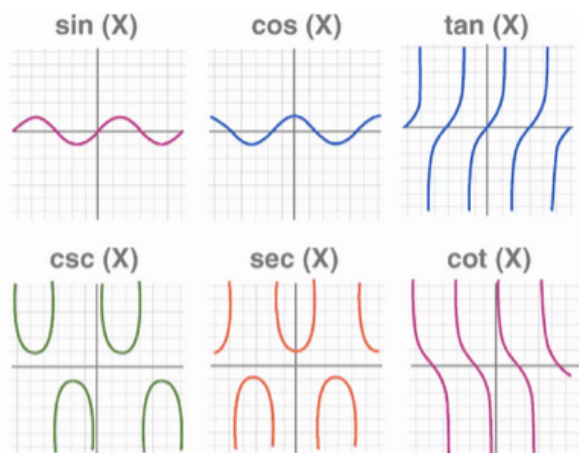
Logarithmic functions are inverse of exponential functions and the methods of plotting them are similar. To plot logarithmic graphs, it is required to make a table of values and then plot the points accordingly on a graph paper. The graph of any log function will be the inverse of an exponential function. An example is given below for better understanding.

For example, the inverse graph of $y=3^x$ will be $y=\log_3\{x\}$ which will be as follows:



Trigonometric Graphs

Trigonometry graphs are plotted below for the 6 trigonometric functions, which include sine function, cosine function, tangent function, cotangent function, cosec function, and sec function. Visit [trigonometry graphs](#) to learn the graphs of each of the functions in detail along with their maximum and minimum values and solved examples.



Frequency Distribution Graph

A frequency distribution graph is used to show the frequency of the outcomes in a particular sample. For frequency distribution graphs, the table of values made by placing the outcomes in one column and the number of times they appear (i.e. frequency) in the other column. This table is known as the frequency distribution table from which the cumulative frequency graph or ogive can be plotted.

There are two commonly used frequency graphs which include:

- Frequency Polygon
- Cumulative Frequency Distribution Graphs

A frequency polygon is almost identical to a histogram, which is used to compare sets of data or to display a cumulative frequency distribution. It uses a line graph to represent quantitative data.

Statistics deals with the collection of data and information for a particular purpose. The tabulation of each run for each ball in cricket gives the statistics of the game. Tables, graphs, pie-charts, bar graphs, histograms, polygons etc. are used to represent statistical data pictorially.

Frequency polygons are a visually substantial method of representing quantitative data and its frequencies. Let us discuss how to represent a frequency polygon.

Steps to Draw Frequency Polygon

To draw frequency polygons, first we need to draw histogram and then follow the below steps:

Step 1- Choose the class interval and mark the values on the horizontal axes

Step 2- Mark the mid value of each interval on the horizontal axes.

Step 3- Mark the frequency of the class on the vertical axes.

Step 4-Corresponding to the frequency of each class interval, mark a point at the height in the middle of the class interval

Step 5-Connect these points using the line segment.

Step 6-The obtained representation is a frequency polygon.

Let us consider an example to understand this in a better way.

Example

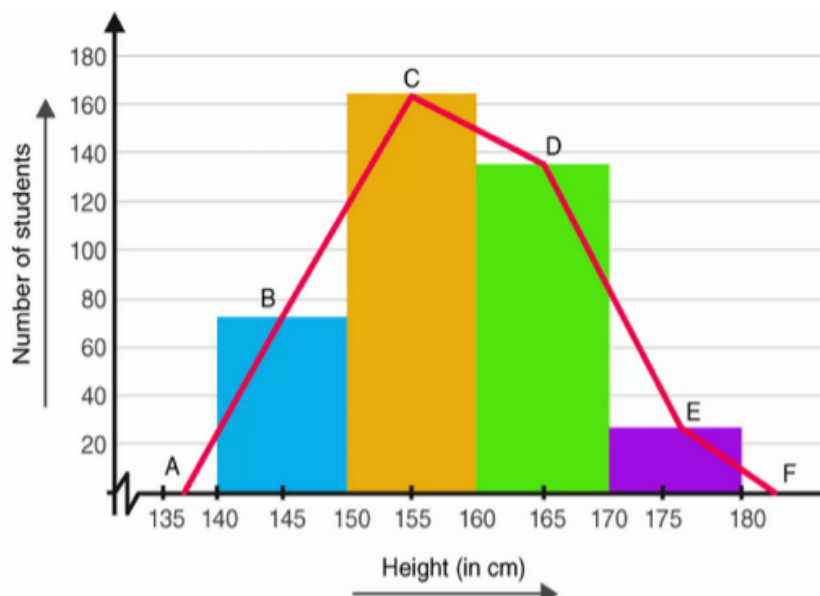
Example 1: In a batch of 400 students, the height of students is given in the following table. Represent it through a frequency polygon.

Height (in cm)	Number of Students(Frequency)
140 – 150	74
150 – 160	163
160 – 170	135
170 – 180	28
Total	400

Solution: Following steps are to be followed to construct a histogram from the given data:

- The heights are represented on the horizontal axes on a suitable scale as shown.
- The number of students is represented on the vertical axes on a suitable scale as shown.
- Now rectangular bars of widths equal to the class size and the length of the bars corresponding to a frequency of the class interval are drawn.

ABCDEF represents the given data graphically in form of frequency polygon as:



Frequency polygons can also be drawn independently without drawing histograms. For this, the midpoints of the class intervals known as class marks are used to plot the points

$$\text{Class Mark} = \frac{\text{Upper Limit} + \text{Lower Limit}}{2}$$

Method Of Moments

What is the Method of Moments?

The method of moments is a way to estimate population parameters, like the population mean or the population standard deviation. The basic idea is that you take known facts about the population, and extend those ideas to a sample.

The method of moments is a statistical technique used for estimating the parameters of a probability distribution based on a sample of data. The basic idea behind the method of moments is to equate the sample moments (such as mean, variance, etc.) with their corresponding population moments, which are expressed in terms of the unknown parameters of the distribution. By solving these equations, we can obtain estimates for the unknown parameters.

Here's a general outline of how the method of moments works:

Choose a probability distribution: First, you need to make an assumption about the underlying probability distribution that generated the data. This assumption is typically based on prior knowledge or the characteristics of the data.

Calculate sample moments: Next, you calculate the sample moments from your data. For example, the first moment is the sample mean, the second moment is the sample variance, and so on.

Equate sample and population moments: Set up equations by equating the sample moments with their corresponding population moments, using the estimates for the moments obtained from the data in step 2 and the expressions for population moments from step 3.

Solve for parameter estimates: Solve the equations from step 4 to obtain estimates for the unknown parameters of the distribution. These estimates are known as the method of moments estimators.

Advantages and Disadvantages

Advantages

- Method of moments is simple (compared to other methods like the maximum likelihood method) and can be performed by hand.

Disadvantages

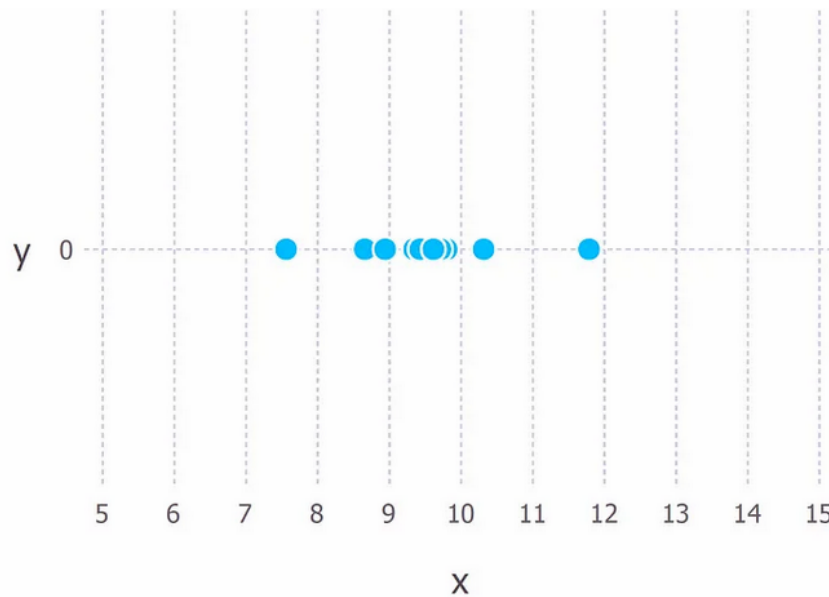
- The parameter estimates may be inaccurate. This is more frequent with smaller samples and less common with large samples.
- The method may not result in sufficient statistics. In other words, it may not take into account all of the relevant information in the sample.

Maximum likelihood estimation

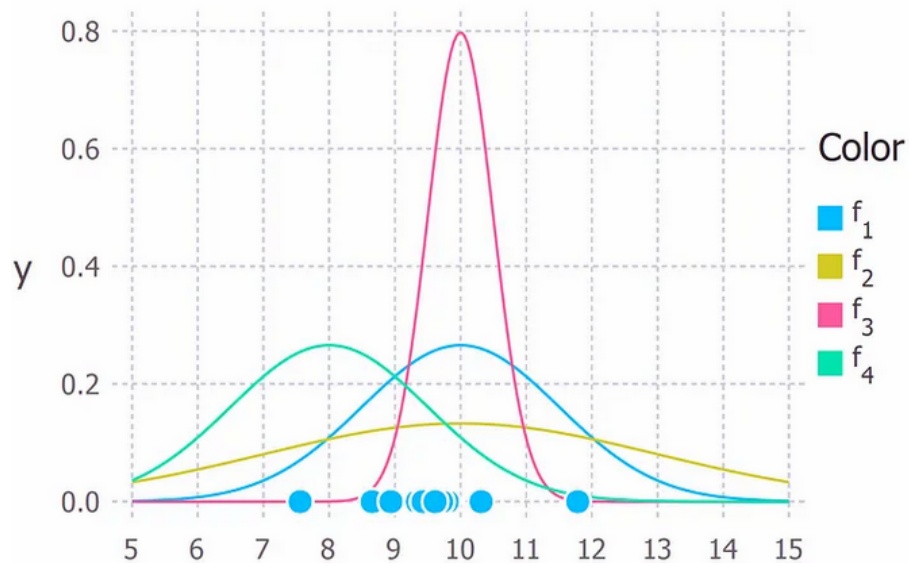
Maximum likelihood estimation is a method that determines values for the parameters of a model. The parameter values are found such that they maximize the likelihood that the process described by the model produced the data that were actually observed.

The above definition may still sound a little cryptic so let's go through an example to help understand this.

Let's suppose we have observed 10 data points from some process. For example, each data point could represent the length of time in seconds that it takes a student to answer a specific exam question. These 10 data points are shown in the figure below



The 10 (hypothetical) data points that we have observed



The 10 data points and possible Gaussian distributions from which the data were drawn. f_1 is normally distributed with mean 10 and variance 2.25 (variance is equal to the square of the standard deviation), this is also denoted $f_1 \sim N(10, 2.25)$. $f_2 \sim N(10, 9)$, $f_3 \sim N(10, 0.25)$ and $f_4 \sim N(8, 2.25)$. The goal of maximum likelihood is to find the parameter values that give the distribution that maximizes the probability of observing the data.

The true distribution from which the data were generated was $f_1 \sim N(10, 2.25)$, which is the blue curve in the figure above.

Session No. 5

Data Modeling Concepts

What is Data Modeling?

Data modeling in analysis is the process of creating a visual representation, abstraction of data structures, relationships, and rules within a system or organization. Determining and analyzing the data requirements required to support business activities within the bounds of related information systems in organizations is another process known as data modeling.

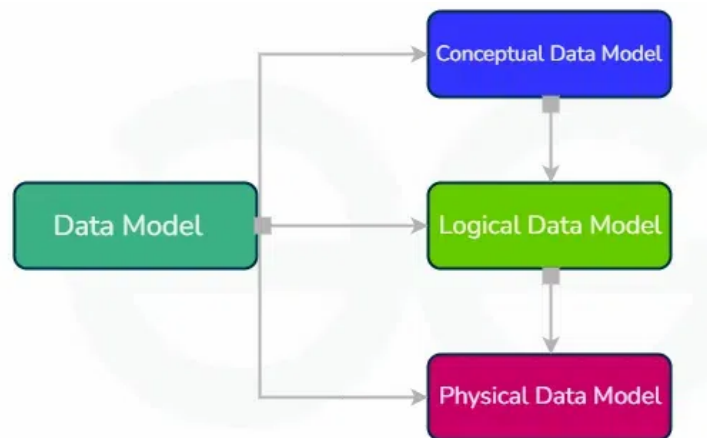
The main objective of data modeling is to provide a precise and well-organized framework for data organization and representation, since it enables efficient analysis and decision-making. Analysts can discover trends, understand the connections between various data items, and make sure that data is efficiently and accurately stored by building models.

What is a Data Model?

Data models are visual representations of an enterprise's data elements and the connections between them. Models assist to define and arrange data in the context of key business processes, hence facilitating the creation of successful information systems. They let business and technical personnel collaborate on how data will be kept, accessed, shared, updated, and utilized within an organization.

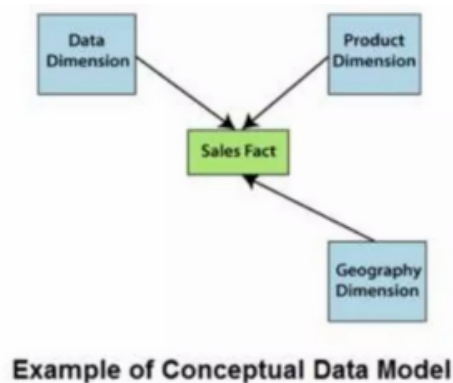
Types of Data Models

There are three main types of data models:



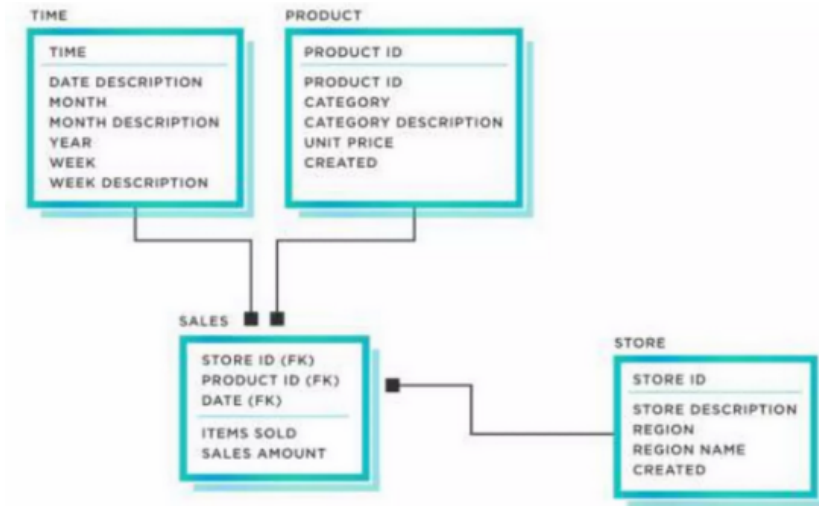
Types of Data Models: Conceptual Data Model

Conceptual Data Model is the first in types of Data Model. They're also known as Domain Models, and they provide a big picture view of the system's contents, how it'll be organized, and which business rules will be involved. Elements are of items that are typically included.



Types of Data Models: Logical Data Model

The Logical Data Model is the second one in the types of Data models. They are less abstract and provide more information on the concepts and relationships in the domain. It follows one of the formal Data Modelling notation systems. Logical data models show the relationships between entities as well as Data properties such as Data types and their respective lengths.



Types of Data Models: Physical Data Model

Physical Data Model is the last one in the types of Data models. They define the format in which data will be physically stored in a database. As a result, they are the least abstract of the bunch.

Data Modeling Process

Data Modeling as a discipline invites stakeholders to examine Data processing and storage in great depth. Different conventions govern which symbols are used to represent data, how models are built up, and how business needs are communicated in Data Modeling methodologies. All approaches provide structured workflows that comprise a list of actions that must be completed in sequential order. An example of a workflow is shown below:

- Determine the entities. Determine the most important properties of each entity. One or more distinct traits called attributes, separate each entity type from the others. For example, a "customer" object might have a first name, last name, phone number, and salutation, but an "address" entity might have a street name and number, as well as a city, state, nation, and zip code.
- Determine the relationships that exist between entities. In the first draft of a Data model, the nature of each entity's relationships with the others will be established. In the above example, each client "resides" at a certain address. If the model was expanded to include an object named "orders," each order would be shipped to and paid to an address. These links are often defined using a single modeling language (UML).
- Complete the attribute-to-entity mapping. This ensures that the model appropriately reflects the company's data usage intentions. There are a variety of formal Data modeling paradigms in use. Object-oriented developers commonly employ analysis and design patterns, although stakeholders from other business areas may utilize different patterns.
- Assign keys as appropriate, and choose a level of normalization that strikes a balance between the need to decrease redundancy and the necessity for performance. Normalization is a method of structuring the various types of Data models (and the databases they represent) in which numerical identifiers, known as keys, are allocated to groupings of data to indicate relationships between them without having to repeat the data. For example, if each client is given a key, that key can be linked to their address and order history without having to duplicate the information in the customer database.

Normalization reduces the amount of storage space required by a database, but at the expense of query performance.

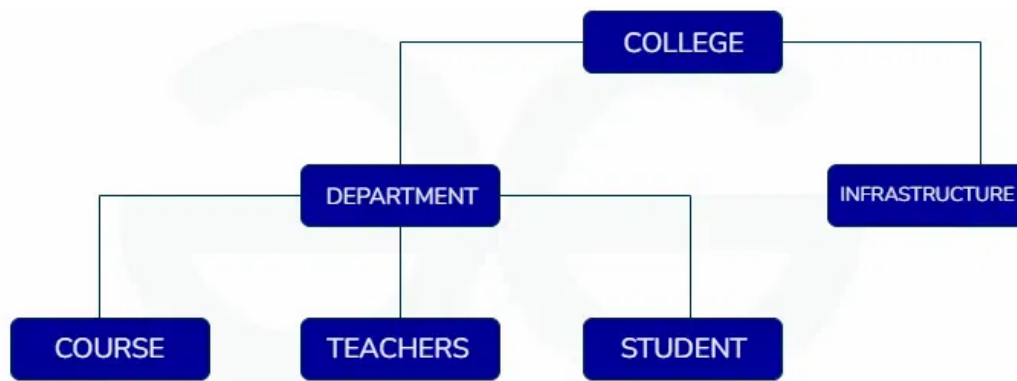
- Complete and test the different types of Data models. As business demands evolve, Data modeling is an iterative process that should be repeated and updated.

Types of Data Modeling

These are the 5 different types of data models:

Hierarchical Model: The structure of the hierarchical model resembles a tree. The remaining child nodes are arranged in a certain sequence, and there is only one root node—or, alternatively, one parent node. However, the hierarchical approach is no longer widely applied. approach connections in the actual world may be modelled using this approach.

For example, in a college there are many courses, many professors and students. So college became a parent and professors and students became its children.



Relational Model :Relational Model represents the links between tables by representing data as rows and columns in tables. It is frequently used in database design and is strongly related to relational database management systems (RDBMS).

Object-Oriented Data Model: In this model, data is represented as objects, similar to those used in object-oriented programming ,Creating objects with stored values is the object-oriented method. In addition to allowing data abstraction, inheritance, and encapsulation, the object-oriented architecture facilitates communication.

Network Model :We have a versatile approach to represent objects and the relationships among these things thanks to the network model. One of its features is a schema, which is a graph representation of the data. An item is stored within a node, and the relationship between them is represented as an edge. This allows them to generalize the maintenance of many parent and child records.

ER-Model: A high-level relational model called the entity-relationship model (ER model) is used to specify the data pieces and relationships between the entities in a system. This conceptual design gives us an easier-to-understand perspective on the facts. An entity-relationship diagram, which is made up of entities, attributes, and relationships, is used in this model to depict the whole database.

A relationship between entities is called an association. Mapping cardinality many associations like:

- one to one
- one to many
- many to one
- many to many

Session No. 6

Understand and model subtypes and supertypes

In data modeling, the concepts of subtypes and supertypes are used to represent relationships between entities and to model hierarchical structures in the data. These concepts are often used in entity-relationship diagrams (ER diagrams) to visually depict the relationships between different types of entities.

Supertypes:

A supertype is a generalization of one or more related entity types. It represents a higher-level entity that encompasses common attributes and relationships shared by its subtypes. Supertypes do not have instances on their own but serve as a template for their subtypes. In other words, a supertype defines a common set of properties and relationships that are inherited by its subtypes.

For example, consider an entity called "Vehicle" as a supertype. It could have attributes like "Manufacturer," "Model," and "Year," as well as relationships like "Owned By" and "Registered In." The "Vehicle" supertype would be more abstract and generalized, representing common features shared by different types of vehicles, such as cars, motorcycles, and trucks.

Subtypes:

A subtype is a specialized entity type that inherits attributes and relationships from its supertype. It represents a more specific category or subclass of the supertype, focusing on distinct characteristics that are unique to that particular subtype.

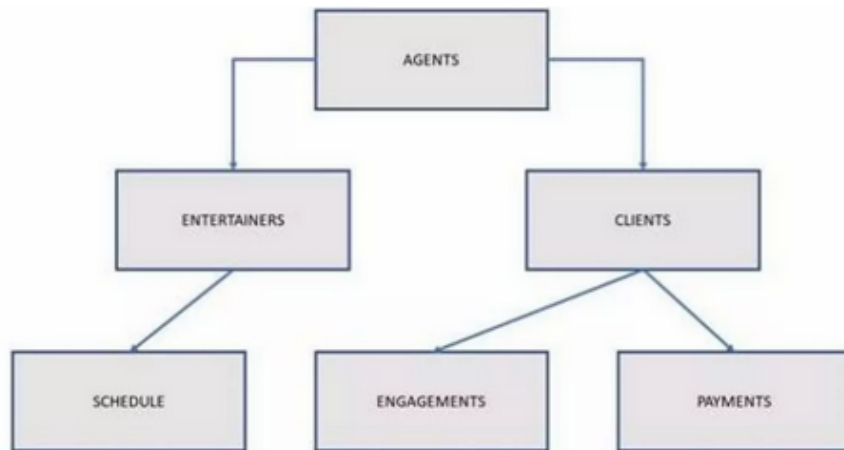
Continuing with the "Vehicle" example, the subtypes could be "Car," "Motorcycle," and "Truck." Each of these subtypes would have its own unique attributes and relationships, such as "Number of Doors" for cars, "Number of Wheels" for motorcycles, and "Cargo Capacity" for trucks.

Understand and model hierarchical data

A hierarchical model represents the data in a tree-like structure in which there is a single parent for each node. To maintain order there is a sort field which keeps sibling nodes into a recorded order. These types of model are designed basically for the early mainframe database management systems, like the Information Management System (IMS) by IBM.

This model structure allows the one-to-one and a one-to-many relationship between two various types of data. This structure is very helpful in describing many relationships in the real world: table of contents, any nested and sorted informatic

Data in this type of database is structured hierarchically and is typically developed as an inverted tree. The "root" in the structure is a single table in the database and other tables act as the branches flowing from the root. The diagram below shows a typical hierarchical database structure



Advantages

- A user can retrieve data very quickly due to the presence of explicit links between the table structures.
- The referential integrity is built in and automatically enforced due to which a record in a child table must be linked to an existing record in a parent table, along with that if a record is deleted in the parent table then that will cause all associated records in the child table to be deleted as well.

Disadvantages

- When a user needs to store a record in a child table that is currently unrelated to any record in a parent table, it gets difficult to record and the user must record an additional entry in the parent table.
- This type of database cannot support complex relationships, and there is also a problem of redundancy, which can result in producing inaccurate information due to the inconsistent recording of data at various sites.

Hierarchical Database Model

A hierarchical database model is a data model in which the data are organized into a tree-like structure. Hierarchical model is one of the oldest database models, dating from the 1960's. First Hierarchical database information management system was developed jointly by North American Rockwell company and IIBM.

Hierarchy is an ordered tree and easy to understand. Data is represented in the form of records. Relationship among the data is represented by records of links. A tree may be defined as a set of nodes. At the root of the tree is the single parent, the parent can have none, one or more children.

Features Of Hierarchical Database Model:

- One to many relationships. It only supports one-to-many relationships. Many to many relationships are not supported.
- Problem in Deletion: If a parent is deleted then the child automatically gets deleted.
- Hierarchy of data: Data is represented in a hierarchical tree-like structure
- Parent-child relationship. Each child can have only one parent but a parent can have more than one child.
- Pointer: Pointers are used for linking records that tell which is a parent and which child record is.
- Disk input and output is minimized. Parent and child records are placed or stored close to each other on the storage device which minimizes the hard disk input and output.
- Fast navigation: As parent and child are stored close to each other so access time is reduced and navigation becomes faster.
- Predefined relationship: All relations between root, parent and child nodes are predefined in the database schema.
- Re-organization difficulty: Hierarchy prevents the re-organization of data.
- Redundancy. One to many relationships increases redundancy in the data which leads to the retrieval of inaccurate data

Hierarchical Model

Advantages

- Promotes data sharing
- Parenchik relationship promotes conceptual simplicity and data integrity
- Database security is provided and enforced by DBMS.
- Efficient with 1:M relationships

Disadvantages

- Requires knowledge of physical da storage characteristics
- Navigational system requires knowledge of hierarchical path
- Changes in structure require changes in all application programs
- Implementation limitations
- No data definition
- Lack of standards

Understand and model recursive relationships

Recursive Relationship in ER Diagrams

ER Diagrams

ER Diagram stands for Entity Relationship Diagram. When we draw the relationships between entities using a diagram, then it is called an entity relationship diagram. The ER diagram is just for understanding the purpose of the database administrator. We cannot use the ER diagram

directly on the computer. ER Diagrams are converted into the tabular form then they are inserted into the computer using any query language. In ER Diagrams, we use attributes, entities, and the relationships between entities. We use an oval shape to represent the entity and a diamond shape to represent the relationships between entities.

Cardinality

The number of the relationship between two entity sets can have, is known as Cardinality. The quantity of items in a particular set is referred to mathematically as its Cardinality. Cardinality can be used by Database Administrators to count values and tables. Cardinality in a Database typically illustrates the relationship between the data in two distinct tables by emphasizing how frequently one object occurs in comparison to another.

The Mapping Cardinality, also known as the Cardinality Ratio, in a database refers to the number of entities that can be connected to another entity through a specific relation set. Although they can help with the description of relation sets with more than two entity sets, Mapping Cardinality is most beneficial when defining binary relation sets.

Types of Cardinality in DBMS:

- One to one cardinality
- Many to one cardinality
- One to many cardinality
- Many to many cardinality

Recursive Relationship

If there is a relationship between two entities and they both belong to the same entity type, then this is called a recursive relationship. We can say a relationship between the same entity set is called a recursive relationship. In the ER diagram, we can use one entity that will enter and exit from itself will denote the recursive relationship. While mapping the ER diagram into the table, we must consider the foreign key in the table for mapping the recursive relationship.

There can be a lot of examples of recursive relationships.

For Example, if a boy marries a girl and both belong to the human entity set, then this is called the recursive relationship.

So, we can indirectly say that if an entity participates more than once in a relationship for different roles, then this is called a recursive relationship. In the above example, the person entity will be used for a boy, and then, on the other hand, it will be used for a girl, and there will be a relation between them.

Real Time Application

Example 1:

One person can have many children, and the relationship between person and child is unary because they both belong to the same entity type as humans.

Example 2:

One monitor can handle multiple students in class, and he is itself a student. So the relationship handling is a unary or recursive relationship because the monitor and other students belong to the same entity set.

Understand and model historical data

What is historical data?

Historical data, in a broad contest, is data collected about past events and circumstances pertaining to a particular subject

By definition, historical data includes most data generated either manually or automatically within an enterprise. Sources, among a great number of possibilities, include press releases, log Oles, financial reports, project and product documentation and email and other communication